

vate arc. These are listed in [Table 83, “Matter-specific DN Object Identifiers”](#). These OID values are assigned by the Connectivity Standards Alliance for use with Matter. All of these Matter-specific RDNs encode values normatively defined as scalars.

When used in Matter Operational Certificate (TLV) format (see [Section 6.5, “Operational Certificate Encoding”](#)), Matter-specific DN attribute types SHALL be encoded in Matter TLV as unsigned integers with the specified length.

When used in X.509 ASN.1 DER format certificate encoding, Matter-specific DN attribute types SHALL have their value encoded as either a [UTF8String](#) or [PrintableString](#) according to the table below. The values SHALL be encoded in network byte order as exactly twice their specified maximum octet length, encoded as uppercase hexadecimal number format without any separators or prefix, and without omitting any leading zeroes.

For example:

- A scalar value [0x0123\\_4567\\_89AB\\_CDEF](#) for [matter-node-id](#):
  - Scalar maximal length: 8 octets (64 bits)
  - Resulting string: "0123456789ABCDEF" (without quotes)
  - Resulting length: 16 characters
- A scalar value [0xAA\\_33CC](#) for [matter-noc-cat](#):
  - Scalar maximal length: 4 octets (32 bits)
  - Resulting string: "00AA33CC" (without quotes)
  - Resulting length: 8 characters

*Table 83. Matter-specific DN Object Identifiers*

| TLV Tag | Matter name                                | Length (octets) | String length | ASN.1 OID                              | Types Allowed in X.509 |
|---------|--------------------------------------------|-----------------|---------------|----------------------------------------|------------------------|
| 17      | <a href="#">matter-node-id</a>             | 8               | 16            | <a href="#">1.3.6.1.4.1.3724.4.1.1</a> | UTF8String             |
| 18      | <a href="#">matter-firmware-signing-id</a> | 8               | 16            | <a href="#">1.3.6.1.4.1.3724.4.1.2</a> | UTF8String             |
| 19      | <a href="#">matter-icac-id</a>             | 8               | 16            | <a href="#">1.3.6.1.4.1.3724.4.1.3</a> | UTF8String             |
| 20      | <a href="#">matter-rcac-id</a>             | 8               | 16            | <a href="#">1.3.6.1.4.1.3724.4.1.4</a> | UTF8String             |
| 21      | <a href="#">matter-fabric-id</a>           | 8               | 16            | <a href="#">1.3.6.1.4.1.3724.4.1.5</a> | UTF8String             |
| 22      | <a href="#">matter-noc-cat</a>             | 4               | 8             | <a href="#">1.3.6.1.4.1.3724.4.1.6</a> | UTF8String             |
| 23      | <a href="#">matter-vvs-id</a>              | 8               | 16            | <a href="#">1.3.6.1.4.1.3724.4.1.7</a> | UTF8String             |